

Macro I & II – 2011-2012 – Lecture 3

Asset Pricing in General Equilibrium

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Changes from version 1.0 are in red

Disclaimer

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0.Introduction

- In this lecture, I want to
 1. show how to compute asset prices in general equilibrium
 2. discuss of the some quantitative properties of asset prices in (simple) GE models

1. Asset Prices in General Equilibrium with Endowments

- I describe here the competitive equilibrium of a pure exchange infinite horizon economy with stochastic Markov endowments.
- This is a basic setting for studying risk sharing, asset pricing, consumption.
- 2 different market structures
 1. Arrow-Debreu structure with complete markets in dated contingent claims all traded at period 0
 2. recursive structure with complete one-period Arrow securities.
- The 2 have different asset structures but identical consumption allocations

1.1. Preferences and endowments

- $\pi(s'|s)$ is a Markov chain with initial distribution $\pi_0(s)$
- $Prob(s_{t+1} = s' | s_t = s) = \pi(s'|s)$ and $Prob(s_0 = s) = \pi_0(s)$
- a sequence of probability measures $\pi(s^t)$ on histories $s^t = [s_t, s_{t-1}, \dots, s_0]$ is given by

$$\pi(s^t) = \pi(s_t | s_{t-1}) \pi(s_{t-1} | s_{t-2}) \dots \pi(s_1 | s_0) \pi_0(s_0) \quad (1)$$

and conditional probability is given by

$$\pi(s^t | s_0) = \pi(s_t | s_{t-1}) \pi(s_{t-1} | s_{t-2}) \dots \pi(s_1 | s_0) \quad (2)$$

1.1. Preferences and endowments

- Trading occurs after s_0 has been observed.
- the probability of state (history) s^t conditional on being in state (history) s^τ at date τ is

$$\pi(s^t|s^\tau) = \pi(s_t|s_{t-1})\pi(s_{t-1}|s_{t-2})\dots\pi(s_{\tau+1}|s_\tau) \quad (3)$$

- (because of Markov property, $\pi(s^t|s^\tau)$ does not depend on history $s^{\tau-1}$)

1.1. Preferences and endowments

- Households : $i = 1, \dots, I$. Each owns a stochastic endowment of one good $y_t^i = y^i(s_t)$, and s_t is publicly observable
- Each household purchase a history-dependant consumption plan $c^i = \{c_t^i(s^t)\}_{t=0}^{\infty}$
- Household objective

$$U(c^i) = \sum_{t=0}^{\infty} \sum_{s^t} \beta^t u[c_t^i(s^t)] \pi(s^t | s_0) = E_0 \sum_{t=0}^{\infty} \beta^t u[c_t^i(s^t)] \quad (4)$$

- u has all nice properties, including $\lim_{c \downarrow 0} u'(c) = +\infty$

1.2. Complete markets

- Household trade dated state-contingent claims to consumption
- $q_t^0(s^t)$ = price of a claim on time-t consumption, contingent on history s^t , in terms of a **numéraire** not specified
- the BC is

$$\sum_{t=0}^{\infty} \sum_{s^t} q_t^0(s^t) c_t^i(s^t) = \sum_{t=0}^{\infty} \sum_{s^t} q_t^0(s^t) y^i(s_t) \quad (5)$$

- Hh problem : choose c^i to maximize (4) s.t. (5)
- Notice that one can collapse the problem into a problem with a single budget constraint because of complete markets

1.2. Complete markets

- let μ^i be the Lagrange multiplier of this constraint, FOC :

$$\frac{\partial U(c^i)}{\partial c_t^i(s^t)} = \mu^i q_t^0(s^t) \quad (6)$$

and with the specification (4) of preferences,

$$\frac{\partial U(c^i)}{\partial c_t^i(s^t)} = \beta^t u'[c_t^i(s^t)] \pi(s^t | s_0) \quad (7)$$

one gets

$$\beta^t u'[c_t^i(s^t)] \pi(s^t | s_0) = \mu^i q_t^0(s^t) \quad (8)$$

1.2. Complete markets

Definition

A **price system** is a sequence of functions $\{q_t^0(s^t)\}_{t=0}^{\infty}$. An **allocation** is a list of sequences of functions $\{c_t^i(s^t)\}_{t=0}^{\infty}$, one for each i . A **feasible allocation** satisfies

$$\sum_i y^i(s_t) \geq \sum_i c_t^i(s^t) \quad (9)$$

Definition

A **competitive equilibrium** is a feasible allocation and price system such that the allocation solves each household problem

1.2. Complete markets

- Notice that (8) implies

$$\frac{u'[c_t^i(s^t)]}{u'[c_t^j(s^t)]} = \frac{\mu^i}{\mu^j} \quad (10)$$

which means that ratios of marginal utilities between pairs of agents are constant across all states and dates.

- An equilibrium allocation solves (10), (9), and (5). Note that (10) implies

$$c_t^i(s^t) = u'^{-1} \left\{ u'[c_t^1(s^t)] \frac{\mu^i}{\mu^1} \right\} \quad (11)$$

and substituting into feasibility condition (9) at equality gives

$$\sum_i u'^{-1} \left\{ u'[c_t^1(s^t)] \frac{\mu^i}{\mu^1} \right\} = \sum_i y^i(s_t) \quad (12)$$

1.2. Complete markets

$$\sum_i u'^{-1} \left\{ u' [c_t^1(s^t)] \frac{\mu^i}{\mu^1} \right\} = \sum_i y^i(s_t) \quad (12)$$

- the RHS of (12) does not depend on the entire history s^t , but only on current state s_t , therefore the LHS, therefore $c_t^1(s^t)$. Then, from (11), it is also the case for all $c_t^i(s^t)$. One then has the following proposition

Proposition

The competitive equilibrium allocation is not history dependent ;
 $c_t^i(s^t) = \bar{c}^i(s_t)$

1.3. Equilibrium pricing function

- Let c^i , $i = 1, \dots, I$ be an equilibrium allocation. Then (6) or (8) gives the price system $q_t^0(s^t)$ as a function of the allocation to Hh i , for any i .
- The price system is a stochastic process
- Because the units of the price system are arbitrary, one can normalized one of the multipliers at any positive value. I set $\mu^1 = u'[c^1(s_0)]$, so that $q_0^0(s_0) = 1$, i.e. the price system is in units of time-0 goods. (one has therefore $\mu^i = u'[c^i(s_0)]$ for all i)

1.4. Example : Risk sharing

- suppose $u(c) = (1 - \gamma)^{-1} c^{1-\gamma}$, $\gamma > 0$ (CRRA). Then (10) implies

$$c_t^i = c_t^j \left(\frac{\mu^i}{\mu^j} \right)^{-\frac{1}{\gamma}} \quad (13)$$

- time- t elements of consumption allocations to distinct agents are constant fractions of one another.
- The individual consumption is perfectly correlated with the aggregate endowment or aggregate consumption.
- The fractions assigned to each individual are independent of the realization of s_t .
- There is extensive cross-time cross-state consumption smoothing.

1.4. Asset pricing

Pricing Redundant Assets

- Let $\{d(s_t)\}_{t=0}^{\infty}$ be a stream of claims on time t , state s^t consumption, where $d(s_t)$ is a measurable function of s_t . The price of an asset entitling the owner to this stream must be

$$a_0^0 = \sum_{t=0}^{\infty} \sum_{s^t} q_t^0(s^t) d(s^t) \quad (14)$$

(this can be understood as an arbitrage equation)

1.4. Asset pricing

Riskless Consol

- A riskless consol offers for sure one unit of consumption at each period, i.e. $d_t(s_t) = 1$ for all t and s_t . The price is

$$a_0^0 = \sum_{t=0}^{\infty} \sum_{s^t} q_t^0(s^t)$$

1.4. Asset pricing

Riskless strips

- Consider a sequence of strips of returns on the riskless consol. The time- t strip is the return process $d_\tau = 1$ if $\tau = t \geq 0$, and 0 otherwise. The price of time- t strip at 0 is

$$\sum_{s^t} q_t^0(s^t)$$

1.4. Asset pricing

Tail assets

- Consider the stream of dividends $\{d(s_t)\}_{t \geq 0}$
- For $\tau \geq 1$, suppose that we strip off the first $\tau - 1$ periods of the dividend and want to get the time-0 value of the dividend stream $\{d(s_t)\}_{t \geq \tau}$.
- Let $a_\tau^0(s^\tau)$ be the time-0 price of an asset that entitles the dividend stream $\{d(s_t)\}_{t \geq \tau}$ if history s^τ is realized :

$$a_\tau^0(s^\tau) = \sum_{t \geq \tau} \sum_{\{\tilde{s}^t: \tilde{s}^\tau = s^\tau\}} q_t^0(\tilde{s}^t) d(\tilde{s}^t) \quad (15)$$

- Let us convert this price into units of time τ , state s^τ by dividing by $q_\tau^0(s^\tau)$:

$$a_\tau^\tau(s^\tau) = \frac{a_\tau^0(s^\tau)}{q_\tau^0(s^\tau)} = \sum_{t \geq \tau} \sum_{\{\tilde{s}^t: \tilde{s}^\tau = s^\tau\}} \frac{q_t^0(\tilde{s}^t)}{q_\tau^0(s^\tau)} d(\tilde{s}^t) \quad (16)$$

1.4. Asset pricing

Tail assets

- Notice that for all consumers i

$$\begin{aligned} q_t^\tau(s^t) &= \frac{q_t^0(s^t)}{q_\tau^0(s^\tau)} = \frac{\beta^t u'[c_t^i(s^t)] \pi(s^t)}{\beta^\tau u'[c_\tau^i(s^\tau)] \pi(s^\tau)} \\ &= \beta^{t-\tau} \frac{u'[c_t^i(s^t)]}{u'[c_\tau^i(s^\tau)]} \pi(s^t | s^\tau) \end{aligned} \quad (17)$$

- Here $q_t^\tau(s^t)$ is the price of one unit of consumption delivered at time t , state s^t in terms of the date- τ , state- s^τ consumption good.
- The price at t for the tail asset is

$$a_\tau^\tau(s^\tau) = \sum_{t \geq \tau} \sum_{\{\tilde{s}^t: \tilde{s}^\tau = s^\tau\}} q_t^\tau(\tilde{s}^t) d(\tilde{s}^t) \quad (18)$$

1.4. Asset pricing

Tail assets

- This tail asset formula is useful if one wants to create in a model a time series of equity prices : an equity purchased at time τ entitles the owner to the dividends from time τ forward, and the price is given by (18).
- Note : The relative price is (17) is not history dependent, given Proposition 1. This is stated in the following proposition :

Proposition

The equilibrium price of date- $t \geq 0$, state- s^t consumption good expressed in terms of date τ ($0 \leq \tau \leq t$), state s^τ consumption good is not history dependent : $q_t^\tau(s^t) = q_t^j(\tilde{s}^k)$ for $j, k \geq 0$ such that $t - \tau = k - j$ and $[s_t, s_{t-1}, \dots, s_\tau] = [\tilde{s}_k, \tilde{s}_{k-1}, \dots, \tilde{s}_j]$.

1.4. Asset pricing

Pricing One Period Returns

- The one-period version of equation (17) is

$$q_{\tau+1}^{\tau}(s^{\tau+1}) = \beta \frac{u'(c_{\tau+1}^i)}{u'(c_{\tau}^i)} \pi(s_{\tau+1}|s_{\tau})$$

- The RHS is the one-period *pricing kernel* at time τ .
- The price at time τ in state s^{τ} of a claim to a random payoff $\omega(s_{\tau+1})$ is given, using the pricing kernel, by

$$\begin{aligned} p_{\tau}^{\tau}(s^{\tau}) &= \sum_{s_{\tau+1}} q_{\tau+1}^{\tau}(s^{\tau+1}) \omega(s_{\tau+1}) \\ &= E_{\tau} \left[\beta \frac{u'(c_{\tau+1})}{u'(c_{\tau})} \omega(s_{\tau+1}) \right] \end{aligned} \tag{19}$$

where superscripts i and dependence to s_{τ} have been deleted.

1.4. Asset pricing

Pricing One Period Returns

- Let denote the one-period gross return on the asset by $R_{\tau+1} = \omega(s_{\tau+1})/p_{\tau}^{\tau}(s^{\tau})$. Then, for any asset, equation (19) implies

$$1 = E_{\tau} \left[\beta \frac{u'(c_{\tau+1})}{u'(c_{\tau})} R_{\tau+1} \right] \quad (20)$$

- The term $m_{\tau+1} = \beta \frac{u'(c_{\tau+1})}{u'(c_{\tau})}$ is a *stochastic discount factor*.

Equation (20) can be understood as a restriction on the conditional moments of returns and $m_{\tau+1}$.

- Applying the law of iterated expectations to equation (20), one gets the unconditional moments restriction :

$$1 = E \left[\beta \frac{u'(c_{\tau+1})}{u'(c_{\tau})} R_{\tau+1} \right] \quad (21)$$

1.5. A Recursive Formulation : Arrow Securities

- One introduce another market structure that preserves the equilibrium allocation from our competitive equilibrium. This setting also preserves the one-period asset-pricing formula (19).
- Arrow (1964) : one-period securities are enough to implement complete markets, provided that new one-period markets are reopened for trading each period
- See Ljungqvist and Sargent for a formal proof

2. Quantitative Model : Mehra & Prescott

2.1. Data

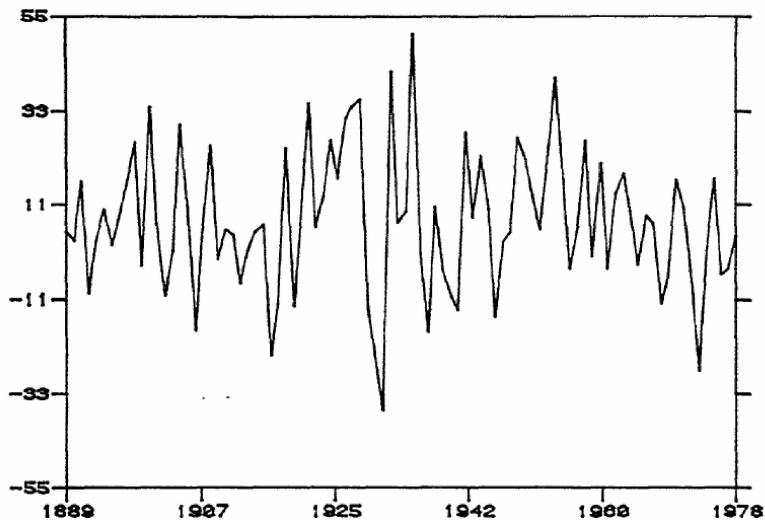


Fig. 1. Real annual return on S&P 500, 1889–1978 (percent).

2. Quantitative Model : Mehra & Prescott

2.1. Data

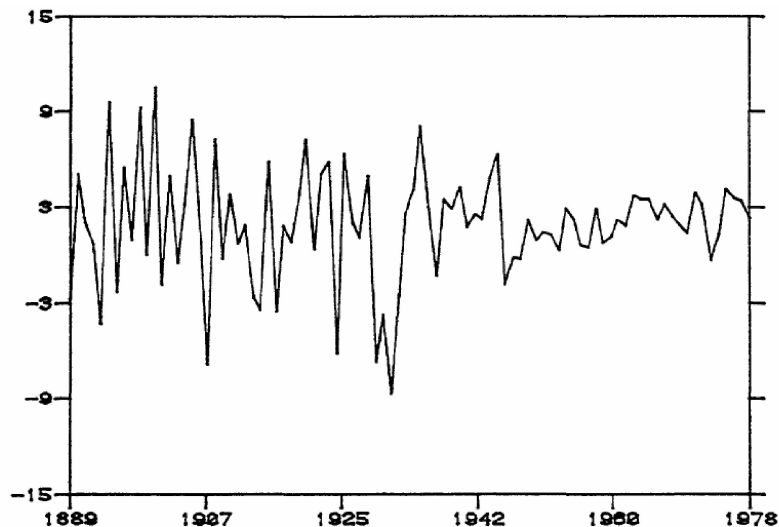


Fig. 2. Growth rate of real per capita consumption, 1889–1978 (percent).

2. Quantitative Model : Mehra & Prescott

2.1. Data

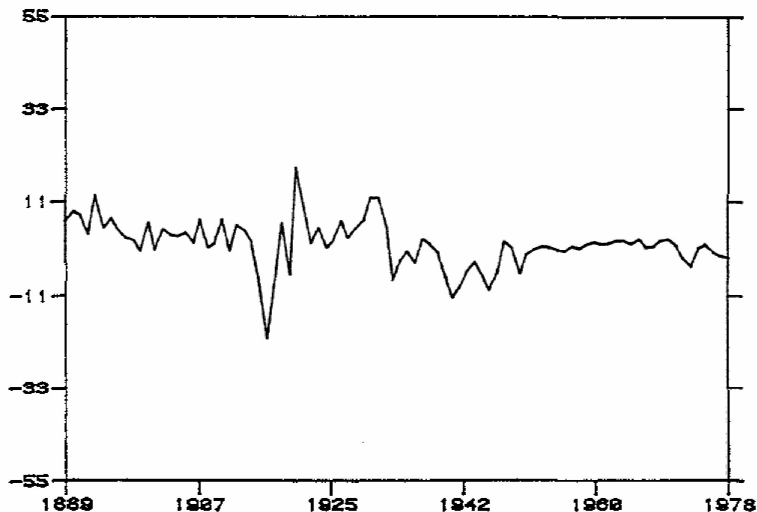


Fig. 3. Real annual return on a relatively riskless security, 1889–1978 (percent).

2. Quantitative Model : Mehra & Prescott

2.1. Data

Table 1

Time periods	% growth rate of per capita real consumption		% real return on a relatively riskless security		% risk premium		% real return on S&P 500	
	Mean	Standard deviation	Mean	Standard deviation	Mean	Standard deviation	Mean	Standard deviation
1889–1978	1.83 (Std error = 0.38)	3.57	0.80 (Std error = 0.60)	5.67	6.18 (Std error = 1.76)	16.67	6.98 (Std error = 1.74)	16.54
1889–1898	2.30	4.90	5.80	3.23	1.78	11.57	7.58	10.02
1899–1908	2.55	5.31	2.62	2.59	5.08	16.86	7.71	17.21
1909–1918	0.44	3.07	-1.63	9.02	1.49	9.18	-0.14	12.81
1919–1928	3.00	3.97	4.30	6.61	14.64	15.94	18.94	16.18
1929–1938	-0.25	5.28	2.39	6.50	0.18	31.63	2.56	27.90
1939–1948	2.19	2.52	-5.82	4.05	8.89	14.23	3.07	14.67
1949–1958	1.48	1.00	-0.81	1.89	18.30	13.20	17.49	13.08
1959–1968	2.37	1.00	1.07	0.64	4.50	10.17	5.58	10.59
1969–1978	2.41	1.40	-0.72	2.06	0.75	11.64	0.03	13.11

2. Quantitative Model : Mehra & Prescott

2.1. Data

- The risk premium is high (6.18 %), as the s.d. of real returns is 5.67% for riskless asset and 16.54% for risky asset.
- Mehra and Prescott have proposed a relatively simple endowment economy to quantitatively reproduce this fact.

2.2. A Pure Exchange Economy

Environment

- representative agent, $E_0 \sum_{t=0}^{\infty} \beta^t u(c_t)$; $0 < \beta < 1$,

$$u(c; \alpha) = \frac{c^{1-\alpha} - 1}{1 - \alpha}$$

- One productive unit (a tree) $\rightsquigarrow$ gives y_t units of a perishable good. This tree is an equity share that is competitively traded, and y_t is its dividend.
- the growth rate of y_t is stochastic : $y_{t+1} = x_{t+1}y_t$
- x is markov : $x_{t+1} \in \{\lambda_1, \dots, \lambda_n\}$, $\text{Prob}(x_{t+1} = \lambda_j | x_t = \lambda_i) = \phi_{ij}$.
It is assumed that this markov chain is ergodic, and that $\lambda_i > 0$, $y_0 > 0$.
- y_t is observed at the beginning of the period and securities are traded ex-dividend.

2.2. A Pure Exchange Economy

Equilibrium

Proposition

Define $A = [a_{ij}]$, $a_{ij} = \beta \phi_{ij} \lambda_j^{1-\alpha}$ and assume $\lim_{m \rightarrow \infty} A^m = 0$. Then, a Debreu competitive equilibrium exists.

2.2. A Pure Exchange Economy

Pricing

- In this economy, the ex dividend price of a security with dividends $\{d_t\}$ is

$$P_t = E_t \left[\sum_{s=t+1}^{\infty} \beta^{s-t} \frac{u'(y_s)}{u'(y_t)} d_s \right]$$

- For the equity, given the functional forms and $d = y$,

$$P_t^e = P^e(y_t, x_t) = E_t \left[\sum_{s=t+1}^{\infty} \beta^{s-t} \frac{y_t^\alpha}{y_s^\alpha} y_s \right]$$

- (y_t, x_t) is a sufficient description of the past history. It defines the state of the economy.

$$P_t^e = E_t \left\{ \beta \frac{u'(y_{t+1})}{u'(y_t)} (P_{t+1}^e + y_{t+1}) \right\}$$

- Given that $y_s = y_t \times x_{t+1} \times x_{t+2} \cdots \times x_s$, P_t^e is homogenous of degree 1 in y_t , which is the current endowment of consumption good.

2.2. A Pure Exchange Economy

Pricing

- Given that equilibrium values of the economy are time invariant functions of (y_t, x_t) , the subscript t can be dropped. The state can be written as (c, i) , where $y_t = c$ and $x_t = \lambda_i$.
- With these notations, the price of an equity satisfies

$$P^e(c, i) = \beta \sum_{j=1}^n \underbrace{\phi_{ij}}_i \underbrace{(\lambda_j c)^{-\alpha}}_{ii} \underbrace{[P^e(\lambda_j c, j) + c \lambda_j]}_{iii} \underbrace{c^\alpha}_{iv}$$

with

- i : probability of state j knowing i
- ii : **marginal utility of tomorrow consumption**
in state j
- iii : tomorrow price + dividend in state j
- iv : **inverse of marginal utility of today consumption**

2.2. A Pure Exchange Economy

Pricing

- Given that P^e is homogenous of degree 1 in c , we can write

$$P^e(c, i) = w_i c$$

where w_i is a constant. Then the pricing equation becomes

$$w_i = \beta \sum_{j=1}^n \phi_{ij} \lambda_j^{(1-\alpha)} (w_j + 1) \quad \forall i = 1, \dots, n$$

- This is a system of n linear equations in n unknowns (the w_i) $\rightsquigarrow$ this has a unique positive solution when a competitive equilibrium exists $\rightsquigarrow$ we can derive prices

2.2. A Pure Exchange Economy

Prices

- The return of an equity if current state is c, i and next state j is

$$r_{ij}^e = \frac{P^e(\lambda_j c, j) + \lambda_j c - P^e(c, i)}{P^e(c, i)} = \frac{\lambda_j (w_j + 1)}{w_i} - 1$$

and the equity expected return is, conditional on state i :

$$R_i^e = \sum_{j=1}^n \phi_{ij} r_{ij}^e$$

- Let us also consider a riskless security that pays 1 unit of good in each state, i.e. $d_j = 1 \forall i$. The price P^f of this asset is

$$P_i^f = P^f(c, i) = \beta \sum_{j=1}^n \phi_{ij} \frac{u'(\lambda_j c)}{u'(c)} d_j = \beta \sum_{j=1}^n \phi_{ij} \lambda_j^{-\alpha}$$

and $R_i^f = 1/P_i^f - 1$

2.2. A Pure Exchange Economy

Prices

- Now we can compute expected returns w.r.t. the stationary distribution
- Let $\pi \in \mathcal{R}^n$ be the vector of stationary probabilities of the markov chain : π is such that

$$\pi = \phi' \pi$$

with $\sum_{i=1}^n \pi_i = 1$ and $\phi = \{\phi_{ij}\}$

- Then we define the expected returns as

$$\begin{aligned} R^e &= \sum_{i=1}^n \pi_i R_i^e \\ R^f &= \sum_{i=1}^n \pi_i R_i^f \end{aligned}$$

and the risk premium is given by $R^e - R^f$.

2.3. Results

- preference parameters : α and β
- Technology parameters $\{\phi_{ij}\}$, $\{\lambda_i\}$: it is assumed that λ takes two values : $\lambda_1 = 1 + \mu + \delta$ and $\lambda_2 = 1 + \mu - \delta$, and $\phi_{11} = \phi_{22} = \phi$, $\phi_{12} = \phi_{21} = 1 - \phi$.
- For US data over 1889-1978, consumption growth = .018, consumption growth s.d. = .036, consumption growth serial correlation = -.14 $\rightsquigarrow \mu = .018$, $\delta = .036$, $\phi = .43$
- Then, we search for (α, β) so that the average risk free rate and the equity risk premium are reproduced.
- α = people's willingness to substitute consumption between successive yearly time period $\rightsquigarrow$ not greater than 10.
- $\beta \in]0, 1[$

2.3. Results

- The model cannot reproduce an equity premium of more than .35%, while it is 6 in the data (given that the risk free rate is .8%)
- There is therefore an **Equity Premium Puzzle**
- A large literature has been devoted to this question
- See Kocherlakota 1996 for a nice survey
- Here I present a quantitative solution of this (quantitative) puzzle, as proposed by German 1998

2.4. A Possible Resolution of the Puzzle

The Model

- Jerman (1998, JME) proposed a model with production, capital, habit formation and K adjustment costs that is quantitatively satisfactory.

- Firms

$$\max E_t \sum_{k=0}^{\infty} \beta^k \frac{\Lambda_{t+k}}{\Lambda_t} [A_{t+k} F(K_{t+k}, X_{t+k} N_{t+k}) - w_{t+k} N_{t+k} - I_{t+k}]$$

where $\beta^k \frac{\Lambda_{t+k}}{\Lambda_t}$ is the MRS of the owners of the firm.

$$K_{t+1} = (1 - \delta)K_t + \phi \left(\frac{I_t}{K_t} \right) K_t$$

- $\phi(\cdot) < 1$ is a positive concave function that models adjustment costs on capital $\rightsquigarrow$ the shadow price of one installed unit of capital, q , differs from the price of one new unit of capital (Tobin's q)
- Firms are financed by retained earnings, and dividends are given by

$$D_t = A_t F(K_t, X_t N_t) - w_t N_t - I_t$$

2.4. A Possible Resolution of the Puzzle

The Model

- Hh

$$\max E_t \sum_{k=0}^{\infty} \beta^k u(c_{t+k})$$

$$\text{s.t. } w_t N_t + a'_t (V_t^a + D_t^a) \geq C_t + a'_{t+1} V_t^a \quad (\Lambda_t)$$

- a_t is a vector of financial assets held at t and chosen at $t - 1$.

this vector contains the representative firm, + possibly other assets. V^a is the vector of asset prices and D^a the vector of dividends payments.

- The Hh also face a time constraint : $N_t + L_t = 1$, and we assume habit persistence : $u = u(c_t - \alpha c_{t-1})$
- Market equilibrium : $A_t F(K_t, X_t N_t) = C_t + I_t$.
- Shocks are to the technology A

2.4. A Possible Resolution of the Puzzle

Model Solution

- If we use a log-linear approximation to solve the model, the expected returns will be the same for all agents $\leadsto$ no possibility to account for the risk premium
- The model is solved by log-linearization, *but* asset prices are computed in a second round using lognormal pricing formulas (see Hansen & Singleton, 1983)
- The model solution can be written

$$s_t = Ms_{t-1} + \varepsilon_t \quad (22)$$

where ε could be a multivariate normal iid shock (In this model, it is univariate).

- Then we use basic asset pricing formula : a claim on future payment $D_{t+k}(s_{t+k})$ has a value

$$V_t(s_t) = \beta^k E_t \left[\frac{\Lambda_{t+k}(s_{t+k}) D_{t+k}(s_{t+k})}{\Lambda_t(s_t)} \right] \quad (23)$$

2.4. A Possible Resolution of the Puzzle

Model Solution

- If Λ and D are lognormal, with distribution given by (22), then the risk free rate can be computed from (23) with $k = 1$ and $D(s_{t+1}) = 1$:

$$E(R_{t,t+1}(s_t)) = \frac{\gamma}{\beta^*} \exp \left\{ \frac{1}{2} (\text{var}(E_t \lambda_{t+1} - \lambda_t) - \text{var}(\lambda_{t+1} - E_t \lambda_{t+1})) \right\}$$

where $\lambda_t = \frac{\Lambda_t}{X_t}$, $X_t = \gamma X_{t-1}$, $\beta^* = \beta \gamma^t$

- The return on equity is given by

$$R_{t,t+1}^d(s_t, s_{t+1}) = \frac{V_{t+1}(s_{t+1}) + D_{t+1}(s_{t+1})}{V_t(s_t)}$$

- German uses simulations to find the unconditional mean.

2.4. A Possible Resolution of the Puzzle

Quantitative predictions

- easy part :
 1. long run restrictions : Cobb-Douglas elasticity on labor = .64, $\gamma = 1.005$ (per quarter), $\delta = .025$
 2. Productivity shocks : A follows a $AR(1)$ process, with persistence .95 or 1, with s.d. of the innovation which is such that postwar US gdp s.d. is reproduced
 3. Risk aversion :

$$\frac{c^{1-\tau}}{1-\tau} \qquad \tau = 5$$

2.4. A Possible Resolution of the Puzzle

Calibration

- Difficult part :
 1. habit formation parameter α
 2. K adjustment cost elasticity ξ
 3. time preference β
 4. shock persistence ρ
- Let $\theta_1 = [\alpha, \beta, \xi, \rho]'$. θ_1 is chosen (estimated) to minimize

$$\mathcal{F} = (\theta_2 - f(\theta_1))' \Omega (\theta_2 - f(\theta_1))$$

where θ_2 is a vector of moments to match, $f(\theta_1)$ is the vector of corresponding moments generated by the model and Ω a weighting matrix. (Simulated Method of Moments)

- $\theta_2 = (\text{s.d. of } c \text{ growth} / \text{s.d. of } y \text{ growth}, \text{ s.d. of } i \text{ growth} / \text{s.d. of } y, \text{ mean risk free rate, equity premium})$, Ω is identity
- The solution is that $\mathcal{F} = .00001$ with

α	ξ	β^*	ρ
.82	.23	.99	.99

2.4. A Possible Resolution of the Puzzle

Quantitative predictions

- The simulation results are

TABLE: Simulation results, Jerman 1998

	$\frac{\sigma_{\Delta c}}{\sigma_{\Delta y}}$	$\frac{\sigma_{\Delta i}}{\sigma_{\Delta y}}$	$E(r^f)$	$E(r^e - r^f)$
Data	.51	2.65	.80	6.18
Standard RBC	.77	1.54	4.26	.02
Standard RBC + $\tau = 10$	.78	1.53	3.36	.26
Habit persistence	.33	3	4.2	.03
K adj. costs	1.14	.68	3.91	.67
Benchmark	.51	2.65	.8	6.18